Stock–Bond Diversification in an Inflation-Driven Rates Regime

Regime-Aware Allocation and Treasury Curve Loss Attribution, 2007–2026

Research Prospectus • Vatsal Maniar • Based on frozen Research Memo v1.1

Research Question & Motivation

Stock–bond diversification rests on the equity–Treasury covariance. That this relationship is regime-dependent, and that nominal bond–stock comovement splits into real-rate and inflation-related components varying over history, is established (Campbell, Pflueger & Viceira, 2026); this project claims neither as novel. It asks: *how does established regime dependence manifest in an implementable portfolio when decisions use only observable, lagged information under realistic costs and constraints, and performance is judged through realized drawdowns and portfolio-level Treasury hedge-book P&L?* Arc: regime detection → implementation → negative benchmark result → 2022 hedge diagnosis → Treasury P&L attribution → prospective cross-episode / out-of-sample work.

Completed Work — Design

Universe. Thirteen liquid ETFs are transparent, reproducible proxies spanning equities (SPY, QQQ, IWM, EFA, EEM), credit (HYG), real estate (VNQ), Treasury duration and cash (TLT, IEF, SHV), TIPS (TIP), gold (GLD), and commodities (DBC). Data span 2007–2026; evaluation is 2008–2026 (lookbacks consume early history, so 2008 is only partly covered).

Signals (observable, lagged). A 90-day SPY–TLT correlation $> +0.20$ flags weakening stock–Treasury diversification; one-month-lagged CPI YoY $> 3\%$ flags elevated inflation; VIX > 25 flags market stress. The active-signal count sets the regime ($0 \rightarrow$ Normal, $1 \rightarrow$ Watch, $\geq 2 \rightarrow$ Defensive) — judgment-based transparent rules, not estimated optima, prioritizing interpretability and traceability.

Implementation. Monthly rebalancing; long-only; a 35% per-asset cap limiting optimizer concentration; 5 bps per unit turnover (a simple linear assumption, not a full execution model); CPI lagged one month to respect publication timing and bar look-ahead; only pre-rebalance information is used.

Benchmarks. Equal Weight, Global Minimum Variance, Maximum Sharpe, SPY, and a fixed-weight 60/40 (SPY/IEF). IEF is a transparent intermediate-duration Treasury proxy for the 40% bond leg — the definition chosen here, not the only possible 60/40.

Key Empirical Results

Strategy	Ann. Ret.	Vol.	Sharpe	Max DD
Regime-Aware	6.0%	10.5%	0.48	−33.3%
60/40	8.4%	11.1%	0.66	−31.9%

The regime-aware rule does not beat 60/40 on full-sample Sharpe (0.48 versus 0.66): detecting deterioration in diversification does not automatically create decision value. The negative result is reported, not reframed as success.

Crisis Evidence

2008 — duration hedged the equity shock; 2020 — the lagged rule was too slow for the crash-and-rebound; 2022 — stock–bond correlation turned positive and added long-duration Treasury exposure worsened the drawdown. In the 2022 overlay test, adding TLT worsened the 60/40 drawdown ≈ 1.0 pp, while DBC (commodities) cut it ≈ 3.5 pp and SHV (cash) ≈ 2.0 pp — moving from *did the hedge fail?* to *why?*

Treasury Attribution (2022)

A second engine reprices a synthetic \$1m long-duration hedge book (modified duration ≈ 9.9, DV01 $\approx$ \$987/bp) under the measured 2022 nominal and real curve move: exact repriced P&L $\approx -$\$191,634 (-19.2%). Channel attribution: real-rate $\approx -$\$278k, convexity $\approx +$\$64k, curve-shape $\approx +$\$37k, breakeven $\approx +$\$11k, residual $\approx -$\$26k. **Within this portfolio-level attribution specification, for this synthetic long-duration book under the measured 2022 replay, the loss is dominated by real-rate repricing rather than the breakeven channel.** This is portfolio P&L attribution, not causal identification — one episode, not evidence that real rates universally dominate Treasury hedge failures. Overlays recover part of the loss (duration-reduction $\approx 28.9\%$, TIPS rotation only $\approx 3.2\%$), motivating, not establishing, that identifying the dominant channel may matter for hedge choice.

Current limits. ETF common history begins in 2007; thresholds are judgment-based and evaluated on the history that motivated them; the 5 bps cost model is simplified; Treasury instruments are synthetic, not CUSIP-level; TIPS treatment is simplified; carry/roll-down is excluded.

Research Prospect

1. Cross-episode P&L attribution. Extend the 2022 framework across historical Treasury hedge failures (longer history, fitted curves, security-level data where feasible), developing a consistent attribution that separates economic yield drivers (real rates, inflation compensation, term premium where identifiable) from curve geometry and pricing effects (level/slope/curvature, convexity, carry/roll) rather than treating overlapping channels as automatically additive. Does the 2022 attribution generalize at all?

2. Macro-regime association. Are macro environments — supply/inflation, demand/growth, monetary-policy, recession/growth-scare, fiscal/term-premium, inflation-expectations — systematically associated with different dominant loss channels? Association, not causal identification, absent a specified design.

3. Out-of-sample identification. Can the dominant channel be identified using only information available as of each decision date, not after the episode? This needs genuine out-of-sample design (pre-specified train/validation/test or rolling-window), with no hindsight-defined states.

4. Decision value. Does mechanism-aware identification improve hedge selection or risk management relative to fixed 60/40, simple duration reduction, or other transparent defensive rules? The arc observation → attribution → real-time identification → better decision? is a hypothesis to test, not a result of Version 1.1.

Research Infrastructure

Both sites are presentation layers over reproducible Python engines — the browsers display, the pipelines compute. **Portfolio Risk & Allocation Analytics** — pra-analytics.netlify.app: the allocation layer (regime classifications, portfolio rules, benchmark comparisons, risk/performance diagnostics, crisis behavior) — the *when/how is the hedge weakening?* side; the Python pipeline, not the browser, is authoritative. **Inflation Regime Rates Risk Engine / YieldShock** — yieldshock.netlify.app: the rates-risk / P&L layer (curve analytics, duration, convexity, DV01/key-rate, measured and stylized shocks, the 2022 attribution, overlays, Vasicek/CIR scenario risk). It exists because the allocation study showed TLT worsened the 2022 outcome but could not explain the rates-risk channel — moving from *the Treasury hedge lost money* to *which rate-risk channels generated that hedge-book P&L?* Two stages of one program: portfolio-lab (behavior / *when*), inflation-lab (attribution / *why*). **Version 1.1 is frozen**; the questions above are the prospective program.